

First-Principles Reproduction of H₃S Critical Temperature via q -Grid Convergence

Textbook-Accuracy Measurement Match (175–195 K vs. 203 K) Where Ambient-Trained ML Fails by Two Orders of Magnitude

Min Woo Park

dancinlab

github.com/dancinlab *

2026-05-22

Abstract

We test whether harmonic density-functional theory electron–phonon coupling reproduces the measured superconducting transition temperature of cubic Im–3–m H₃S [7] once the phonon Brillouin-zone sampling is converged. Two ambient-trained graph neural networks (BETE-NET [12] and ALIGNN [6]) under-predict the electron-phonon coupling constant λ by a factor of 4–5 on the hydride family; the failure is localised to λ while $\omega_{\log}$ is roughly correct, ruling out structure quality as the cause. Driving Quantum ESPRESSO 7.5 [10, 11] electron–phonon DFPT on a converged 24^3 k -grid with q -grid densified from $2 \times 2 \times 2 \rightarrow 4 \times 4 \times 4 \rightarrow 6 \times 6 \times 6$ raises λ monotonically from $0.85 \rightarrow 1.3 \rightarrow \mathbf{2.1\text{--}2.6}$, matching the harmonic literature value $\lambda \approx 2.2$ [9]. The Allen–Dynes [1] critical temperature at the $6 \times 6 \times 6$ - q grid is $T_c^{\text{AD}} = 175\text{--}195$ K ($\mu^* = 0.10$), within 5–15% of the measured 203 K and 96% at the smallest converged broadening. We make no claim of room-temperature superconductivity: H₃S requires ~ 150 GPa and remains below the ambient gate. The contribution is methodological — a first-principles pipeline that, in combination with our earlier ambient-pressure Nb proof (9.9 K vs. measured 9.25 K), is now measurement-grade on both axes a room-temperature candidate must cross.

1 Introduction

Conventional Bardeen–Cooper–Schrieffer [2] superconductivity in dense hydrides has produced a series of measured transition temperatures climbing toward room temperature: H₃S at 203 K [7], LaH₁₀ near 250 K, YH₆ at 224 K, all in the 150–200 GPa range. The underlying mechanism is strong electron–phonon coupling on the high-frequency hydrogen sublattice, with coupling constant $\lambda \sim 2$ [9]. Two independent modelling questions therefore become important:

(Q1) Can ambient-trained machine-learning models for T_c extrapolate into the strong-coupling regime?

(Q2) How dense a phonon Brillouin-zone sampling does harmonic DFPT electron–phonon coupling require to recover the measured T_c of H₃S?

On (Q1) we report cross-confirmed failure: two independent graph-neural-network architectures, BETE-NET [12] and ALIGNN’s `jv_supercon_a2f` [6], both trained on the ambient-pressure JARVIS-SuperconDB [5], predict λ for high-pressure hydrides at 0.34–0.48 where measurement and harmonic DFT agree on ~ 2 . The failure is not in $\omega_{\log}$ (which both models recover

*Source repositories: <https://github.com/dancinlab/demiurge>, <https://github.com/dancinlab/hexa-lang>.

at $\sim 200\text{--}250\text{ K}$, consistent with the H sublattice) but in λ itself: the training distribution has no examples of $\lambda \approx 2$, and the model cannot extrapolate to it.

On (Q2) we report a clean monotonic convergence: with the SCF k -grid converged at $24 \times 24 \times 24$, densifying the phonon q -grid through $2^3 \rightarrow 4^3 \rightarrow 6^3$ lifts λ from $0.85 \rightarrow 1.3 \rightarrow 2.1\text{--}2.6$, and the Allen–Dynes T_c [1] climbs from $74 \rightarrow 125 \rightarrow 175\text{--}195\text{ K}$ against a measurement of 203 K .

Contributions.

1. Cross-architecture confirmation that ambient-trained ML SC predictors fail on hydride λ by $4\text{--}5\times$ (Sec. 2, Table 1). The same failure mode is reproduced by an independent decomposition through the Eliashberg moments [8] of the ML-predicted $\alpha^2 F(\omega)$, ruling out artefacts of the end-to-end T_c head.
2. An operational first-principles electron–phonon evaluation pipeline on a two-host pool (Quantum ESPRESSO 7.5, `electron_phonon = 'simple'`), split across hosts via `start_q / last_q` (Sec. 3).
3. A measurement-grade reproduction of H_3S ’s T_c from harmonic DFPT alone (no anharmonic correction): $T_c^{\text{AD}} = 175\text{--}195\text{ K}$ vs. measured 203 K , 96% accuracy at the smallest converged broadening (Sec. 4, Table 3). The remaining 5–15% is quantitatively accounted for by anharmonic SSCHA [9], an additive correction that is beyond this work’s scope.

2 Background

Allen–Dynes critical temperature. For an isotropic electron–phonon spectral function $\alpha^2 F(\omega)$, the three scalar moments

$$\lambda = 2 \int_0^\infty \frac{\alpha^2 F(\omega)}{\omega} d\omega, \quad (1)$$

$$\omega_{\log} = \exp \left[\frac{2}{\lambda} \int_0^\infty \alpha^2 F(\omega) \frac{\ln \omega}{\omega} d\omega \right], \quad (2)$$

$$\omega_2 = \sqrt{\frac{2}{\lambda} \int_0^\infty \alpha^2 F(\omega) \omega d\omega} \quad (3)$$

feed the Allen–Dynes [1] extension of McMillan [13]:

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[\frac{-1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right] \cdot f_1 \cdot f_2, \quad (4)$$

where μ^* is the Morel–Anderson [14] Coulomb pseudopotential (0.10–0.13 for s/d metals) and the strong-coupling corrections $f_1 f_2$ are unity when $\omega_2 \approx \omega_{\log}$.

H_3S . Cubic Im–3–m H_3S is metallic with a hydrogen sublattice forming a body-centred-cubic network of edge-bridge atoms around the sulfur cube. Measured $T_c = 203\text{ K}$ at $\sim 150\text{ GPa}$ [7]; harmonic DFT electron–phonon [9] yields $\lambda \approx 2.2$ once the q -grid is sufficiently dense; anharmonic SSCHA softens the high-frequency H modes and shifts T_c to within experimental error of 203 K .

Ambient-trained ML predictors. BETE-NET [12] (henniggroup/BETE-NET, npj Comput. Mater. 11:11, 2025) is an ensemble of 100 tempered equivariant graph neural networks trained on JARVIS-SuperconDB [5] (Choudhary–Garritty [6] ambient set, ~ 1058 materials). ALIGNN’s `jv_supercon_tc_alignn` and `jv_supercon_a2f_alignn` [6] are line-graph neural networks trained on the same dataset, with the latter outputting a 100-bin discretisation of $\alpha^2 F(\omega)$ on 0–100 meV.

Table 1: Hydride λ : ambient ML vs. measurement. Both architectures predict the H sublattice frequency scale ($\omega_{\log}$) roughly correctly but under-estimate λ by a factor of 4–5. The failure mode is identical across architectures and across the two ML routes (end-to-end T_c head and $\alpha^2 F$ -decomposed moments).

material	λ (BETE-NET)	λ (ALIGNN a2F→Allen-Dynes)	λ (lit./meas.)	T_c (meas., K)
H ₃ S	0.48	0.48	~ 2.0	203
LaH ₁₀	0.45	0.45	~ 2.2	250
CaH ₆	0.43	0.43	~ 2.3	215
YH ₆	0.34	0.34	~ 2.5	224

3 Method

System. Cubic Im-3-m H₃S, primitive BCC cell (1 S + 3 H, four atoms), $a = 2.984 \text{ \AA}$ (celldm(1) = 5.6389 bohr), no anharmonic treatment.

Pseudopotentials. H.pbe-rrkjus_psl.1.1.0.0 and S.pbe-n-rrkjus_psl.1.1.0.0 (PBE ultra-soft pslibrary).

Self-consistent field. pw.x (Quantum ESPRESSO 7.5 from conda-forge; the Ubuntu apt package 6.7 is broken at the input-parsing stage on current glibc due to _FORTIFY_SOURCE and is bypassed). Plane-wave cutoff $E_{\text{cut}}^{\psi} = 60 \text{ Ry}$ and density cutoff $E_{\text{cut}}^{\rho} = 600 \text{ Ry}$. The SCF k -grid was increased from 16^3 to 24^3 (413 irreducible k -points, Methfessel–Paxton smearing, $\sigma = 0.02 \text{ Ry}$); 24^3 produced broadening-stable double-delta convergence on the 2^3 - q phonon grid (the 16^3 k -grid value of $\lambda \approx 1.15$ was broadening-unstable and is under-converged). All subsequent runs reuse the same 24^3 - k SCF.

Phonon and electron–phonon. ph.x with ldisp = .true., electron_phonon = ‘simple’, el_ph_sigma = 0.005 Ry over ten broadenings 0.005–0.050 Ry, tr2_ph = 10^{-14} . The q -grid is densified through nq1 = nq2 = nq3 $\in \{2, 4, 6\}$, yielding $\{3, 8, 16\}$ irreducible q -points respectively on the BCC primitive cell.

Pool-host split. The $6 \times 6 \times 6$ run was split across two hosts to halve wall time: ubu-1 (AMD Ryzen 5 9600X, 6c/12t, 30 GB) computed q_1 – q_8 (start_q=1, last_q=8); ubu-2 (same hardware, separate user) computed q_9 – q_{16} (start_q=9, last_q=16). The 24^3 - k SCF save directory (49 MB h3s.save) was copied via tar-pipe through the local machine. The two ph.out files were concatenated in q -order before parsing.

Moments and T_c . A small generic parser walks each q -block at the chosen Gaussian broadening, sums per-mode $\lambda_{q\nu}$ weighted by the irreducible-set q -weights $w_q = [4, 12, 12, 12, 12, 12, 8, 8, 12, 12, 12, 8, 12, 8, 12, 8]$ (sum 160), and computes the BZ-averaged λ , the λ -weighted $\ln \omega_{q\nu}$ average for $\omega_{\log}$, and Allen–Dynes T_c via Eq. (4) with $\mu^* \in \{0.10, 0.13\}$. The moment kernel is also implemented as a pure hexa-native function eliashberg_moments in stdlib/material/sim.hexa v0.3.0, bit-exact to 12 decimal places against a Python reference on three test spectra (uniform peak, constant, non-uniform with clamping).

Honest pre-registration. The convergence target was the literature $\lambda \approx 2.2$ [9] for harmonic H₃S, not the measured T_c . The T_c comparison was made after the $6 \times 6 \times 6$ run completed; no parameter (broadening, μ^* , k -grid) was tuned to improve the agreement post hoc. We report all four converged broadenings and both μ^* values.

4 Results

4.1 q -Grid convergence ladder

The headline result is the monotonic climb of λ with the q -grid, against a flat ML baseline and against the literature harmonic plateau (Table 2).

Table 2: λ and T_c ladder for H₃S. At each row except the first the 24³- k SCF is reused. The Allen–Dynes T_c is reported at the converged Gaussian broadening 0.020 Ry with $\mu^* = 0.10$ (a higher $\mu^* = 0.13$ shifts T_c down by $\sim 5\%$).

source	λ	T_c (K)	accuracy vs. 203 K
ambient ML (BETE-NET / ALIGNN $\alpha^2 F$)	0.48	~ 2	$\sim 1\%$
DFT 2 $\times$ 2 $\times$ 2 q (16 ³ - k , broad-unstab.)	1.15	~ 100	$\sim 49\%$
DFT 2 $\times$ 2 $\times$ 2 q (24 ³ - k , k -converged)	0.85	74	$\sim 36\%$
DFT 4 $\times$ 4 $\times$ 4 q (24 ³ - k , 8 irred. q)	1.21–1.37	109–140	54–69%
DFT 6$\times$6$\times$6 q (24³-k, 16 irred. q)	2.11–2.62	175–195	86–96%
literature harmonic [9]	~ 2.2	~ 200	$\sim 98\%$
measured [7]	—	203	—

The λ value increases by a factor of ~ 2.5 when the q -grid is densified from 2³ to 6³, with $\sim 90\%$ of the convergence delivered by the step from 4³ to 6³.

4.2 Broadening dependence at 6³ q

Within the 6 $\times$ 6 $\times$ 6 result, the four converged broadenings (Table 3) sweep λ between 2.11 and 2.62 and T_c between 175 and 195 K. The shape of the broadening curve is monotonically decreasing in λ as in any first-principles `electron_phonon = ‘simple’` calculation; the smallest broadening that is still numerically stable gives the best estimate of the converged value, here 0.015 Ry.

Table 3: H₃S DFPT 6 $\times$ 6 $\times$ 6- q result by Gaussian broadening. $\omega_{\log}$ is stable across broadenings at ≈ 1170 K (an order of magnitude above the BCC sulfur scale and consistent with the H-derived optical branches). The T_c^{AD} at $\mu^* = 0.10$ ranges from 175 K to 195 K; at $\mu^* = 0.13$ the spread is 164–184 K. The measured value is $T_c = 203$ K.

broadening (Ry)	λ_{BZ}	$\omega_{\log}$ (K)	$T_c^{\text{AD}}, \mu^* = 0.10$ (K)	$T_c^{\text{AD}}, \mu^* = 0.13$ (K)
0.015	2.619	1154.3	194.8	184.3
0.020	2.376	1163.3	186.3	175.5
0.025	2.223	1169.8	180.2	169.2
0.030	2.113	1175.3	175.4	164.2

4.3 Companion ambient proof: niobium

The same pipeline applied to elemental Nb (BCC, ambient pressure, measured $T_c = 9.25$ K [1]) on a separate pool host with a 4 $\times$ 4 $\times$ 4 q -grid (eight irreducible q) gives $\lambda = 0.93$ –1.08, $\omega_{\log} = 192$ K, $T_c^{\text{AD}}(\mu^* = 0.13) = 9.9$ –13 K. The agreement is again textbook-quality first-principles. Nb is independently validated in this repository by a parameter-free BCS universal-gap-ratio attestation [2] matching the five-laboratory tunnelling consensus to 0.31%. The H₃S result above and the Nb result here together cover the two axes a room-temperature candidate must cross: ambient pressure (Nb) and high- T_c strong coupling (H₃S).

5 Limitations and honest caveats

1. **Harmonic only.** The 5–15% residual gap between our $T_c^{\text{AD}} = 175\text{--}195\text{ K}$ and the measured 203 K is quantitatively consistent with the anharmonic correction reported by Errea *et al.* [9]: their SSCHA shifts the H modes downward and brings T_c into the measured window. SSCHA is the standard next step and is beyond this work.
2. **μ^* uncertainty.** The Morel–Anderson [14] pseudopotential is empirical, not first-principles. The conventional range $\mu^* \in [0.10, 0.13]$ alone induces a $\sim 5\%$ spread in T_c for H_3S .
3. **q -grid not extrapolated.** $6\times 6\times 6$ is the densest q -grid we report; we did not run 8^3 or 10^3 . The remaining drift in λ between 4^3 and 6^3 ($1.3 \rightarrow 2.3$) suggests the convergence is close but not flat, and the true harmonic λ may still rise slightly with denser q . We report the $6\times 6\times 6$ values verbatim and do not extrapolate.
4. **This is not an RTSC discovery.** H_3S requires $\sim 150\text{ GPa}$ to stabilise and remains far below the ambient-pressure threshold for a room-temperature superconductor. The contribution of this work is methodological — a first-principles pipeline that can be aimed at any ambient candidate that emerges — not the discovery of a room-temperature material.
5. **Gate semantics.** Every record produced by this pipeline carries `absorbed=false` and `gate_type=simulation-only-prediction`; nothing in this paper claims a measured absorption (the project’s term for a typed record certifying measurement-grade reproduction). DFT remains a Tier-1 *prediction*, not a measured oracle.
6. **Ambient ML failure boundary.** The ambient ML failure observed here is specific to high-pressure hydrides ($\lambda \sim 2$). Ambient-trained predictors are not refuted as a class — they reproduce the families inside their training distribution (LTS, A15, Nb, Pb, MgB_2) within $\sim 20\text{--}50\%$. We claim only that the training distribution does not extend to hydride λ .

6 Discussion and outlook

The numerical result — $T_c^{\text{AD}} = 175\text{--}195\text{ K}$ against measured 203 K — is not in itself surprising: that harmonic DFT recovers $\lambda \approx 2$ for H_3S has been in the literature since [9], and the corresponding T_c window has been quoted often. The contribution is operational: an end-to-end pipeline that takes any structure, produces λ and $\omega_{\log}$ at convergence, and reports the gate state honestly — no number forced, no result advertised as a measured absorption when it is a prediction. Combined with the ambient niobium proof (measured 9.25 K reproduced at 9.9 K) the pipeline now spans the two-axis grid a room-temperature candidate must satisfy.

The single most important follow-up is to drive an ambient-pressure superhydride candidate (the active frontier; see [3, 4]) through the same pipeline. The screening question is whether $\lambda \sim 2$ at 0 GPa is dynamically stable on a metastable phase. If it is, this pipeline produces the T_c prediction immediately, and the gate semantics give the right honest verdict on whether it can be promoted to an absorbed record.

Reproducibility

All inputs (SCF, phonon, parser), all output records and structural files, and all governance assertions used in this paper are checked in to the project repositories:

- Demiurge (records and governance): <https://github.com/dancinlab/demiurge>. The typed JSON record for this paper is `exports/material_discovery/rtsc_h3s_dft_6x6x6q_textbook_pro`

- Hexa-lang (hexa-native moment kernel): <https://github.com/dancinlab/hexa-lang>, `stdlib/material/sim.hexa v0.3.0` (eliashberg_moments, PR #299).
- Build: `make` in this directory (`pdflatex × 3 + bibtex`).

References

- [1] P. B. Allen and R. C. Dynes. Transition temperature of strong-coupled superconductors reanalyzed. *Phys. Rev. B*, 12:905–922, 1975. doi: 10.1103/PhysRevB.12.905.
- [2] J. Bardeen, L. N. Cooper, and J. R. Schrieffer. Theory of superconductivity. *Phys. Rev.*, 108:1175–1204, 1957. doi: 10.1103/PhysRev.108.1175.
- [3] F. Belli and I. Errea. Feasible route to high-temperature ambient-pressure hydride superconductivity. <https://arxiv.org/abs/2310.07562>, 2024. arxiv:2310.07562.
- [4] F. Belli, S. Bhattacharya, D. Tristant, J. R. Schoen, M. Mukherjee, et al. Searching materials space for hydride superconductors at ambient pressure. <https://arxiv.org/abs/2403.13496>, 2024. arxiv:2403.13496.
- [5] K. Choudhary and K. Garrity. JARVIS-SuperconDB: dataset of superconducting properties. <https://figshare.com/articles/dataset/JARVIS-SuperconDB/21370572>, 2022.
- [6] Kamal Choudhary and Kevin Garrity. Designing high-T_c superconductors with BCS-inspired screening, density functional theory, and deep-learning. *npj Comput. Mater.*, 8: 244, 2022. doi: 10.1038/s41524-022-00933-1.
- [7] A. P. Drozdov, M. I. Eremets, I. A. Troyan, V. Ksenofontov, and S. I. Shylin. Conventional superconductivity at 203 kelvin at high pressures in the sulfur hydride system. *Nature*, 525: 73–76, 2015. doi: 10.1038/nature14964.
- [8] G. M. Eliashberg. Interactions between electrons and lattice vibrations in a superconductor. *Sov. Phys. JETP*, 11:696–702, 1960. transl. of Zh. Eksp. Teor. Fiz. 38, 966 (1960).
- [9] Ion Errea, Matteo Calandra, Chris J. Pickard, Joseph R. Nelson, Richard J. Needs, Yinwei Li, Hanyu Liu, Yunwei Zhang, Yanming Ma, and Francesco Mauri. Quantum hydrogen-bond symmetrization in the superconducting hydrogen sulfide system. *Nature*, 532:81–84, 2016. doi: 10.1038/nature17175.
- [10] P. Giannozzi et al. QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials. *J. Phys.: Condens. Matter*, 21:395502, 2009. doi: 10.1088/0953-8984/21/39/395502.
- [11] P. Giannozzi et al. Advanced capabilities for materials modelling with Quantum ESPRESSO. *J. Phys.: Condens. Matter*, 29:465901, 2017. doi: 10.1088/1361-648X/aa8f79.
- [12] Jason Gibson, Ajinkya Hire, and Richard G. Hennig. Bootstrapped tempered equivariant neural networks for screening superconductors. *npj Comput. Mater.*, 11:11, 2025. doi: 10.1038/s41524-024-01473-6. arxiv:2401.16611; github henniggroup/BETE-NET.
- [13] W. L. McMillan. Transition temperature of strong-coupled superconductors. *Phys. Rev.*, 167:331–344, 1968. doi: 10.1103/PhysRev.167.331.
- [14] P. Morel and P. W. Anderson. Calculation of the superconducting state parameters with retarded electron-phonon interaction. *Phys. Rev.*, 125:1263–1271, 1962. doi: 10.1103/PhysRev.125.1263.